

Blockchain-Based Secure Platform for Identity, Access Control & Digital Asset Management

Smart India Hackathon 2026 — Complete Project Reference Document

1. Problem Statement Details

Field	Detail
PS Number	SIH26125
Organization	Bharat Electronics Limited (BEL)
Category	Software
Theme	Blockchain & Cybersecurity
Submission Deadline	20 September 2026

2. Background & Problem

Organizations rely heavily on centralized identity and access management systems, which create security and operational risks — vulnerability to cyberattacks, identity theft, unauthorized access, and single points of failure. Digital and physical asset ownership is often managed through disconnected or semi-centralized systems, making verification of authenticity, access rights, and ownership history difficult and unreliable.

There is a growing need for a decentralized, tamper-proof system that can securely manage user identities, control access permissions, and ensure transparent ownership of digital assets.

3. What the Official PS Asks For

- **Decentralized Identifiers (DIDs)**
 - Each user assigned a secure, verifiable digital identity independent of centralized authorities, authenticated via cryptographic proofs
- **NFT-Based Digital Asset Ownership**
 - Digital assets represented as NFTs — unique, traceable, permanently recorded on-chain
 - NFTs directly allocated to user identities, establishing verifiable, non-duplicable ownership
- **Smart Contract Governance**
 - Only authorized administrators can mint NFTs and assign them to identities
- **Role-Based Access Control (RBAC)**
 - Roles: Admin, Manager, Auditor, User — each with defined access rights
 - Permissions enforced automatically by smart contracts during all operations
- **Immutable Audit Trail**
 - Every identity creation, NFT mint, ownership transfer, and permission update permanently recorded on-chain

4. Unique Features to Differentiate Your Build

Note: DID + NFT + RBAC is a well-established Web3 pattern (Ontology, Verite, HashKey DID, and many academic/open-source implementations already exist). The features below are what would meaningfully differentiate your specific implementation rather than replicate an existing platform.

i. AI-Powered Access Anomaly Detection

Layer behavioral anomaly detection on top of the smart contract rules — flagging when a role (e.g.,



Edit with WPS Office

Auditor) accesses assets outside its normal pattern, even if technically permitted. No major existing DID platform combines blockchain identity with AI-based behavioral monitoring.

ii. Hybrid Emergency Revocation Mechanism

Pure blockchain identity systems struggle with emergency response since records are immutable. A time-locked, multi-signature revocation mechanism lets a quorum of admins freeze/revoke a compromised identity quickly — addressing a real, cited weakness of platforms like Ontology and Verite.

iii. Focused Indian Institutional Use Case

Rather than a generic identity demo, scope tightly to one concrete, underserved problem:

- Anti-fake-degree verification — NFT-backed academic certificates that employers can verify instantly on-chain, addressing India's fake-certificate problem directly
- Government asset/equipment custody chain — tamper-proof tracking of ownership/custody transfers for lab or hospital equipment

iv. DPDP Act (India) Compliance Layer

Existing DID platforms are built for global/Western compliance regimes. Explicitly mapping consent management and data-minimization operations to India's Digital Personal Data Protection Act (DPDP) 2023 is a differentiator few current products offer natively.

v. Cross-Chain Portability

Most enterprise blockchain identity tools lock into a single chain (Ethereum or Hyperledger only). A lightweight abstraction layer allowing identity/NFT records to migrate or verify across multiple chains avoids vendor lock-in — important for government adoption.

5. End-to-End Workflow

- Stage 1 — Identity Onboarding: user registers, system generates a DID using a public-private key pair
- Stage 2 — Cryptographic Verification: identity authenticated via digital signature, no central authority involved
- Stage 3 — Role Assignment: admin assigns a role (Admin / Manager / Auditor / User) to the DID via smart contract
- Stage 4 — Asset Minting: authorized admin mints an NFT representing a digital/physical asset
- Stage 5 — Asset Allocation: NFT is linked to a specific user identity, establishing verifiable ownership
- Stage 6 — Access Enforcement: smart contract automatically checks RBAC rules on every operation request
- Stage 7 — Anomaly Check (your differentiator): AI layer evaluates the access request against historical behavior patterns
- Stage 8 — Immutable Logging: every action (identity creation, mint, transfer, permission change) is recorded on-chain
- Stage 9 — Emergency Response (your differentiator): multi-signature quorum can freeze/revoke a compromised identity
- Stage 10 — Audit & Verification: any party can query the blockchain to verify ownership, authenticity, and access history

6. Tech Stack

Layer	Tools
Blockchain / Smart Contracts	Solidity, Ethereum (testnet) or Hyperledger Fabric
Development Framework	Hardhat or Truffle (compile, test, deploy contracts)
NFT Standard	ERC-721 (unique assets) — ERC-1155 if batching asset types
DID Standard	W3C DID spec; ERC-1056 or ERC-725 for on-chain identity registries



Layer	Tools
Frontend	React.js + Ethers.js / Web3.js for wallet interaction
Wallet Integration	MetaMask for identity signing and transactions
Off-chain Storage	IPFS — for certificate/asset metadata not stored on-chain
AI Anomaly Detection	Python microservice (scikit-learn) monitoring access logs, flags anomalies via API
Backend / API layer	Node.js (Express) or Python (FastAPI) connecting frontend, blockchain, and AI service
Database (off-chain)	MongoDB or PostgreSQL — for indexing/querying on-chain events quickly

7. Realistic Hackathon MVP Scope

- Deploy DID registry + RBAC + NFT minting smart contracts on a testnet (e.g., Sepolia)
- Simple React frontend: register identity, assign role, mint/view NFT-backed certificate or asset
- One working use case demo — e.g., issue and verify a student's NFT-backed degree certificate
- Basic anomaly-flagging demo — a script showing an out-of-pattern access attempt getting flagged
- On-chain audit trail view — simple table pulling event logs from the smart contract

8. Who Uses This Application

- Educational institutions — issuing and verifying tamper-proof academic credentials
- Government departments — managing custody and transfer records of equipment/assets
- Auditors / compliance teams — verifying access history and ownership without relying on a central authority
- Administrators / IT teams — managing roles and access rights across the organization

9. Social Impact

- Reduces fake-degree and credential fraud — a widespread, real problem in Indian education and hiring
- Removes single points of failure in identity systems, reducing risk of mass identity theft from a breached central database
- Gives citizens/employees self-sovereign control over their own identity, rather than trusting one centralized authority
- Improves institutional trust — employers, auditors, and government bodies can independently verify claims without contacting the issuing body

10. Existing Market — What Already Exists

This is the most saturated pattern among the problem statements reviewed. DID + NFT + smart-contract RBAC is a widely studied and implemented Web3 architecture.

Product / Project	Overlap
Ontology (ONT ID)	Full decentralized identity and data management platform, own blockchain, global partnerships
Verite	Identity verification for smart contracts, web apps, mobile apps without exposing personal data
HashKey DID	Cross-platform identity authentication with ~840,000 users
Deloitte Smart ID (open-source)	Attribute-endorsement identity model using smart contracts with RBAC-style permissions
Academic research (Ethereum vs Hyperledger ABAC)	Direct prior comparison of attribute-based access control implementations using smart contracts

Given this, your differentiation must come from the specific use case (e.g., anti-fake-degree verification), the AI anomaly layer, and India-specific compliance mapping — not from the base DID/NFT/RBAC architecture itself, which is largely established.



Edit with WPS Office

11. GitHub Repositories & Reference Code

- [SmartIdentity / smartId-contracts](#) — Deloitte's open-source Smart ID contracts; attribute-endorsement identity model with role-based permissions via smart contracts. Closest real-world reference to your PS.
- [GitHub Topic: smart-contracts \(Solidity\)](#) — broad set of example Solidity smart contract projects, including identity/token patterns
- [GitHub Topic: nft \(Solidity\)](#) — ERC-721 templates and example NFT minting/marketplace contracts
- [GitHub Topic: solidity-smart-contracts](#) — includes soulbound (non-transferable) NFT identity/reputation examples — relevant for degree-certificate use case
- [panaverse / defi-dapps-solidity-smart-contracts](#) — teaching repo covering NFT minting, wallet integration, testnet deployment end-to-end
- [crazyrabbitLTC / nft-contracts](#) — Hardhat + TypeChain + Ethers NFT contract template, good project -structure starting point

12. Standards & Reference Documents

- [W3C Decentralized Identifiers \(DID\) spec](#) — official technical standard for DIDs
- [ERC-721 Standard](#) — official NFT standard reference
- [ERC-1056 \(Lightweight Identity\)](#) — on-chain identity registry standard used by several DID implementations
- [Digital Personal Data Protection Act 2023 \(India\)](#) — for your compliance-mapping differentiator, search official text on meity.gov.in

13. Judge Q&A Preparation — What's Genuinely New

- Lead with the specific use case (e.g., anti-fake-degree verification) — not generic 'blockchain identity'
- Emphasize the AI anomaly detection layer — this is the clearest technical differentiator from existing DID platforms
- Be upfront that DID/NFT/RBAC is an established pattern, and clearly state your specific contribution on top of it — honesty here is more credible than overclaiming novelty
- Have the DPDP Act compliance angle ready if asked about data protection — most competitors don't address Indian law specifically

Compiled from SIH 2026 official problem statement data and project discussion notes.



Edit with WPS Office